

Stage 1: Multi-Source Literature Retrieval and Deduplication

We retrieve papers from three complementary academic databases to maximize coverage and enable cross-source deduplication. The retrieval window begins at 2005, encompassing the period when Energy-Based Model and variational Bayesian research (dayan1995, helmholtz, lecun2006) provided mathematical precursors to what Friston formalized as the Free Energy Principle in 2006 (friston2006); this inclusive start captures historical lineage and cross-disciplinary convergence that a later cutoff would exclude.

arXiv. We query the arXiv Atom API using phrase-matched searches including `all:"active inference"`, `all:"free energy principle"`, `all:"expected free energy"`, `all:"variational free energy" AND all:"inference"`, and targeted Energy-Based Model queries (`all:"energy-based model" AND all:"free energy"`, `all:"Helmholtz machine" AND all:"inference"`, `all:"Boltzmann machine" AND`